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PROJECT AVATAR

Trace the Pattern,
Target the Cure:
Advancing Public Health
Through Data Science

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EXECUTIVE SUMMARY

The Philippines experiences frequent tropical cyclones, severe flood inundation, diesel-driven ambient air pollution, and surface urban heat extremes that disproportionately impact vulnerable urban communities. Project "Avatar" introduces an integrated environmental health surveillance framework that pairs multi-source geospatial data with tabular machine learning and a lightweight Large Language Model (LLM) powered by Retrieval-Augmented Generation (RAG). We combine satellite observations (Sentinel-5P, Landsat, MODIS) and hydrometeorological products (PAGASA, SRTM DEM) with gridded demographic records from the Philippine Statistics Authority and WorldPop. Specialized machine learning models downscale ambient fine particulate matter $PM_{2.5}$ to a 1 km grid, map surface urban heat islands, and forecast lagged flood-associated leptospirosis outbreaks. The lightweight LLM queries an authoritative RAG knowledge base of World Health Organization (WHO) and local health guidelines to translate complex multi-hazard risk indices into proactive, plain-language advisories. By unifying the environmental domains of air, heat, water, and terrain, the system provides local government units and residents with anticipatory, actionable public health protection.

KEYWORDS

Environmental Health, Spatiotemporal Machine Learning, Multi-Hazard Early Warning System, Particulate Matter, Surface Urban Heat Island, Hydrometeorological Disease Forecasting, Retrieval-Augmented Generation (RAG)

BACKGROUND

The Philippines ranks among the most disaster-prone nations globally, facing recurrent tropical cyclones, extreme flooding, urban air pollution, and climate-induced heatwaves (Tito et al., 2024).

Environmental exposures directly impact population health outcomes across four distinct vectors in the Philippines:

1. **Air quality:** Heavy transit congestion and diesel public utility vehicle (PUV) emissions generate elevated fine particulate matter (Collado et al., 2022).
2. **Extreme heat:** Temperature anomalies and severe El Niño dry spells increase thermal stress, escalating cardiovascular emergencies and heat-related deaths (Vergara & Blanco, 2023).
3. **Water:** Heavy rainfall and flooding contaminates drinking water, disrupts sanitation, and increases the risk of waterborne diseases (Tan et al., 2025) .
4. **Landfill**: Soil moisture depletion and soil erosion happens when there is too much heat or precipitation in the country (Jaranilla-Sanchez et al., 2011).

THE PROBLEM

In the Philippines, poor waste management, air pollution, extreme heat, and unsafe drinking water are interconnected environmental challenges that can worsen one another and compound public health risks. However, their combined impacts remain difficult to assess, limiting the ability to identify vulnerable communities and develop effective, coordinated interventions.

OBJECTIVES

- **Specific:** Develop a location-aware mobile health application that sends real-time notifications and personalized hazard alerts to citizens based on four environmental domains
- **Measurable:** Assess all four environmental indicators and generate a composite risk score to identify and rank high-risk areas
- **Attainable:** Utilize publicly available datasets, incorporating risk assessment, visualization, and location-based notification features
- **Relevant:** Provide the general public with actionable health advisories
- **Time-bound:** Complete the assessment framework and prototype within the competition's timeframe

SCOPE & LIMITATIONS

Scope: Focuses on metropolitan and peri-urban centers in the Philippines, prioritizing high-density regions such as Metro Manila. Integrates satellite remote sensing (Sentinel, Landsat, MODIS), topographic surfaces, and Philippine Statistics Authority (PSA) / WorldPop demographics. Employs tabular machine learning for hazard estimation and a lightweight LLM + RAG system strictly for synthesizing hazard metrics into public health advisories.

Limitations: Ground air-quality validation is constrained by the limited number and spatial sparsity of DENR-EMB reference stations. Epidemiological validation relies on municipal-level reporting cadences, which are subject to local reporting lags. The lightweight LLM is strictly decoupled from raw spatial raster calculations to eliminate mathematical hallucinations, operating solely over precomputed model indices.

METHODOLOGY

Multi-Source Data Ingestion & Preprocessing:

- Atmospheric layers (Sentinel-5P TROPOMI trace gases, MODIS/VIIRS Aerosol Optical Depth) and meteorological variables (ERA5-Land) are ingested alongside OpenStreetMap road network data.
- Land Surface Temperature (LST) and Normalized Difference Built-up Index (NDBI) are extracted from Landsat 8/9 and MODIS imagery.
- Hydrologic indicators, soil moisture proxies, and topographic variables are derived from PAGASA rainfall advisories, satellite precipitation, and the SRTM Digital Elevation Model.
- Temporal lag features ranging from 1 to 8 weeks are computed to capture floodwater stagnation and pathogen incubation windows.

Machine Learning Hazard Modeling:

- An Extreme Gradient Boosting (XGBoost) or spatial random forest model downscales ambient $PM_{2.5}$ to a 1 km × 1 km spatial grid.
- An unsupervised clustering algorithm (HDBSCAN or Gaussian Mixture Models) computes heat-health vulnerability indices using the IPCC risk formula ($Risk = Hazard \times Exposure \times Vulnerabil$) based on elderly populations and informal settlement density.
- A LightGBM model paired with SHAP (SHapley Additive exPlanations) values predicts multi-week lagged municipal leptospirosis risk thresholds.

Retrieval-Augmented Generation (RAG) & Light LLM Synthesis:

- Risk outputs are linked to an indexed knowledge base containing WHO AirQ+ health risk frameworks, Department of Health advisories, and disaster risk reduction protocols.
- A lightweight LLM extracts context-specific clinical guidelines matching the computed risk severity to generate proactive, localized warnings for schools, transit workers, and municipal health officers.

Dashboard & Notification Interface:

- The system presents exposure heatmaps, high-risk transit corridors, and automated advisories through an interactive web-based dashboard (Streamlit/Dash).

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